Next-Generation Rib Fracture Diagnosis: Vision-Language Model (VLM) for Grounded Report Generation & Patient-Readable Radiology

Current Limitations

- ✗ Binary classification (fracture / no fracture)
- ✗ No fracture localization
- ✗ No interpretable report generation
- ✗ No patient-facing output
- ✗ Ignores free-text clinical notes
- ✗ Post-hoc Grad-CAM only
- ✗ Single binary output → limited clinical utility

Key Innovations

✓ Trimodal Input
CXR + structured EHR + free-text notes

✓ Fracture-Specific Grounding
Anatomical rib priors + per-rib detection

✓ Dual-Audience Generation
Clinical report + patient summary

✓ Visual Question Answering (VQA)
Clinician queries → grounded responses

Proposed Vision-Language Model (VLM) Architecture

Input Modalities (Trimodal)

Chest X-Ray
Medical Vision Transformer
(BiomedCLIP / ViT)

Structured EHR
Feed-Forward Network
(26 clinical variables)

Free-Text Clinical Notes
ClinicalBERT / GatorTron
(triage, nursing, physician notes)

Cross-Attention Multimodal Fusion

Medical LLM Decoder
Fine-tuned LLaMA-3 / Mistral (LoRA)

Three Simultaneous Outputs

① Grounded Fracture Localization
Bounding boxes / segmentation masks
Per-rib detection with spatial grounding

② Structured Clinical Report
Radiologist-style findings
Clinical impression & recommendations

③ Patient-Readable Summary (6th-grade reading level)
Plain-language translation of findings
Optimized for comprehension (RLHF + readability rewards)

Expected Improvements Over Current Pipeline

Output Type	Localization	Input Modalities	Interpretability	Patient Comms	CDS Integration
Binary prob → **Grounded report + VQA**	Grad-CAM → **Bounding boxes / masks**	2 modalities → **Trimodal (CXR + EHR + notes)**	Post-hoc → **Inherent (grounded text)**	None → **Auto-generated plain-language**	Binary alert → **Full report + discharge education**

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